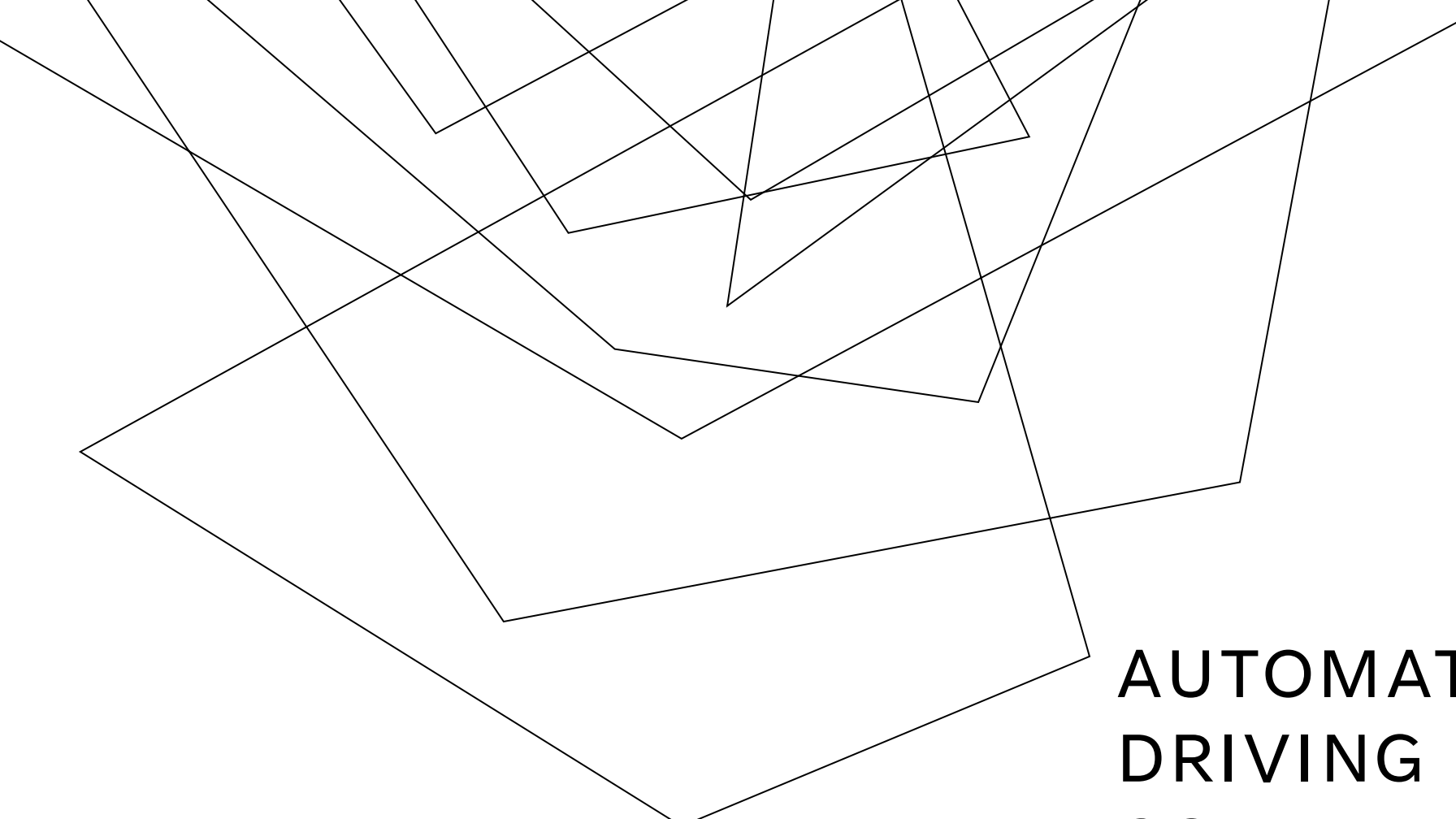


A series of thin, black, overlapping lines forming various geometric shapes like triangles and polygons, creating a complex, abstract pattern in the upper left portion of the slide.

AUTOMATIC DRIVING COMPUTING

3. PERCEPTION (1)





지난 시간 복습

SENSORS

- 1. Role of sensors in autonomous driving cars**
- 2. Types of sensors in autonomous driving cars**
 - Camera
 - LiDAR
 - RADAR
- 3. Case study**



CF) RADAR IN SPACE

SAR (Synthetic Aperture Radar) 위성





목차

목차

1. Perception – overview
2. Image classification
3. Convolutional neural network
4. Vision transformer

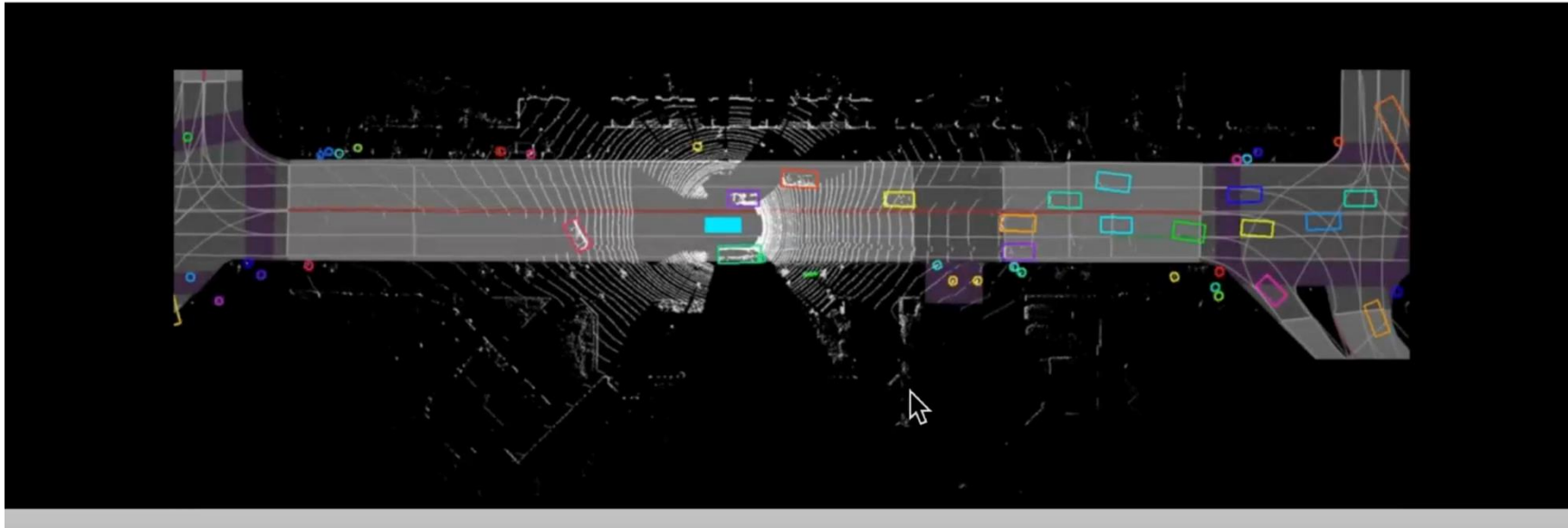


An abstract graphic design featuring two thin, dark grey lines that intersect on a light grey background. One line is oriented diagonally from the top-left towards the bottom-right, while the other is oriented from the top-right towards the bottom-left. The intersection point is located to the left of the text.

PERCEPTION - OVERVIEW

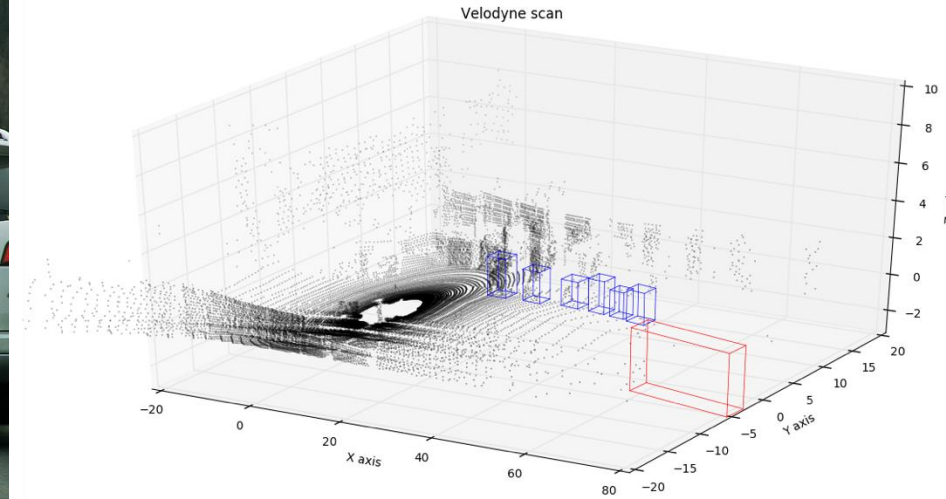
TECH STACKS OF AUTONOMOUS DRIVING

Autonomy Stack - Perception



PERCEPTION

Sensor 데이터를 가지고 주위의 물체들을 어떻게 식별 (Perception) 할까?



PERCEPTION

Sensor 데이터를 가지고 주위의 물체들을 어떻게 식별 (Perception) 할까?

1) Camera (image) 데이터 먼저



PERCEPTION

Sensor 데이터를 가지고 주위의 물체들을 어떻게 식별 (Perception) 할까?

- 1) Camera (image) 데이터 먼저
- 2) 단일 물체만 포함되어 있는 이미지 먼저 (= Image classification)
(Divide and conquer!)



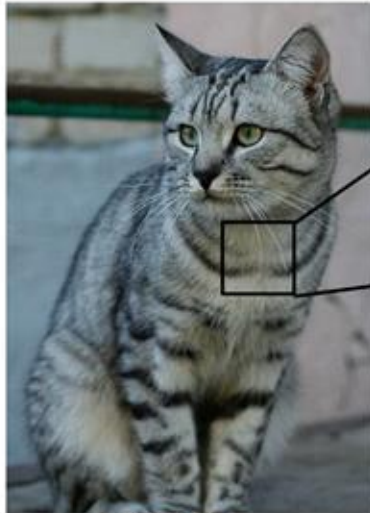
An abstract graphic consisting of two thin, dark gray lines that intersect diagonally on a light gray background. One line slopes downwards from left to right, while the other slopes upwards from left to right.

IMAGE CLASSIFICATION

IMAGE CLASSIFICATION

Input : image tensor

The Problem: Semantic Gap



This image by Nikita is
licensed under [CC-BY 2.0](#)

```
[[105 112 100 111 104 99 106 99 96 103 112 119 104 97 93 87]
 [ 91 98 102 106 104 79 98 103 99 105 123 136 110 105 94 85]
 [ 76 85 98 105 128 105 87 96 95 99 115 112 106 103 99 85]
 [ 99 81 81 93 120 131 127 100 95 98 102 99 96 93 101 94]
 [106 91 61 64 69 91 88 85 101 107 109 98 75 84 96 95]
 [114 100 85 55 55 69 64 54 64 87 112 129 98 74 84 91]
 [133 137 147 103 65 81 80 65 52 54 74 84 102 93 85 82]
 [128 137 144 140 109 95 86 70 62 65 63 63 68 73 86 101]
 [125 133 148 137 119 121 117 94 65 79 80 65 54 64 72 98]
 [127 125 131 147 133 127 126 131 111 90 89 75 61 64 72 84]
 [115 114 109 123 150 148 131 118 113 109 100 92 74 65 72 78]
 [ 89 93 98 97 100 147 131 118 113 114 113 109 106 95 77 80]
 [ 63 77 86 81 77 79 102 123 117 115 117 125 125 130 115 87]
 [ 62 65 82 89 70 71 80 101 124 126 119 101 107 114 131 119]
 [ 63 65 75 88 89 71 62 81 120 130 135 105 81 90 110 110]
 [ 87 65 71 87 106 95 69 45 76 130 126 107 92 94 105 112]
 [110 97 82 86 117 123 116 66 41 51 95 93 89 95 102 107]
 [164 146 112 88 82 120 124 104 76 48 45 66 88 101 102 109]
 [157 170 157 120 93 86 114 132 112 97 69 55 70 82 99 94]
 [130 120 134 161 139 100 109 118 121 134 114 87 65 53 69 86]
 [120 112 96 117 150 144 120 115 104 107 102 93 87 81 72 79]
 [123 107 96 86 83 112 153 149 122 109 104 75 80 107 112 99]
 [122 121 102 80 82 86 94 117 145 140 153 102 50 70 92 107]
 [122 164 140 103 71 56 78 83 93 103 119 139 102 61 69 84]]
```

What the computer sees

An image is a tensor of integers
between [0, 255]:

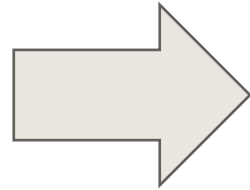
e.g. 800 x 600 x 3
(3 channels RGB)



IMAGE CLASSIFICATION

Output : pre-defined category set 각각에 대한 score

airplane	-2.4
bicycle	-1.1
cat	5.1
dog	3.2
frog	0.7
person	2.9

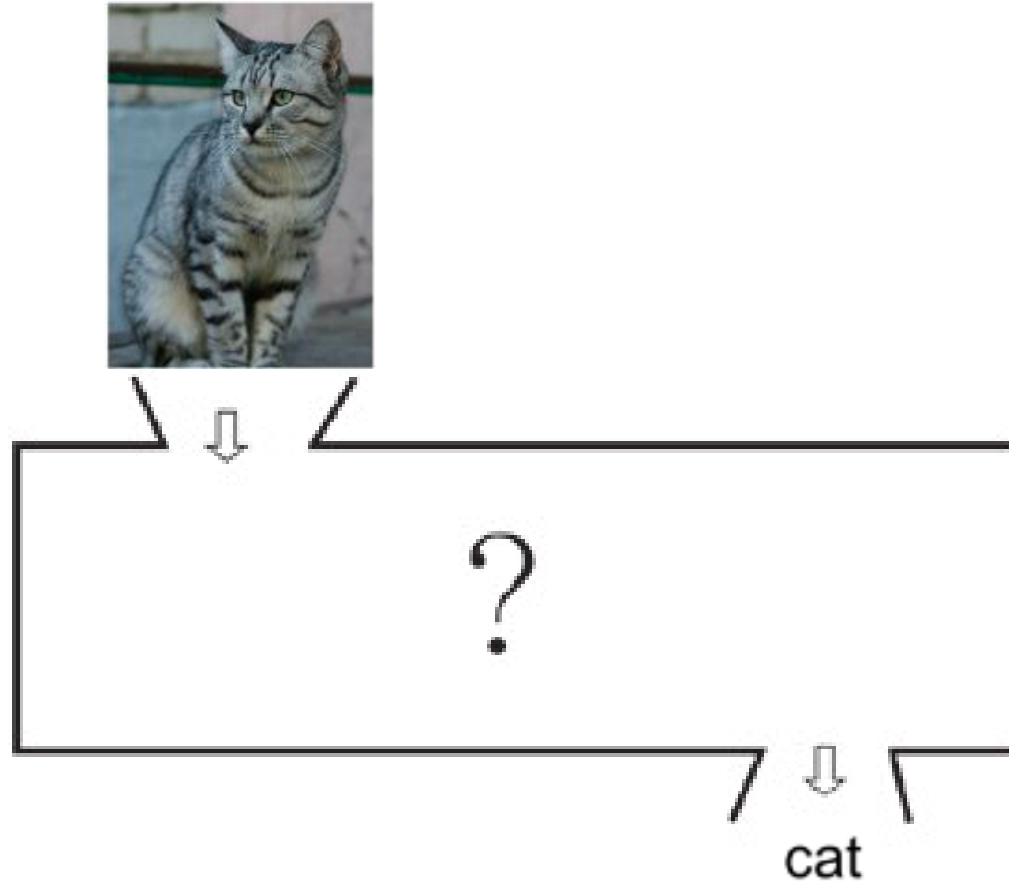


“cat”



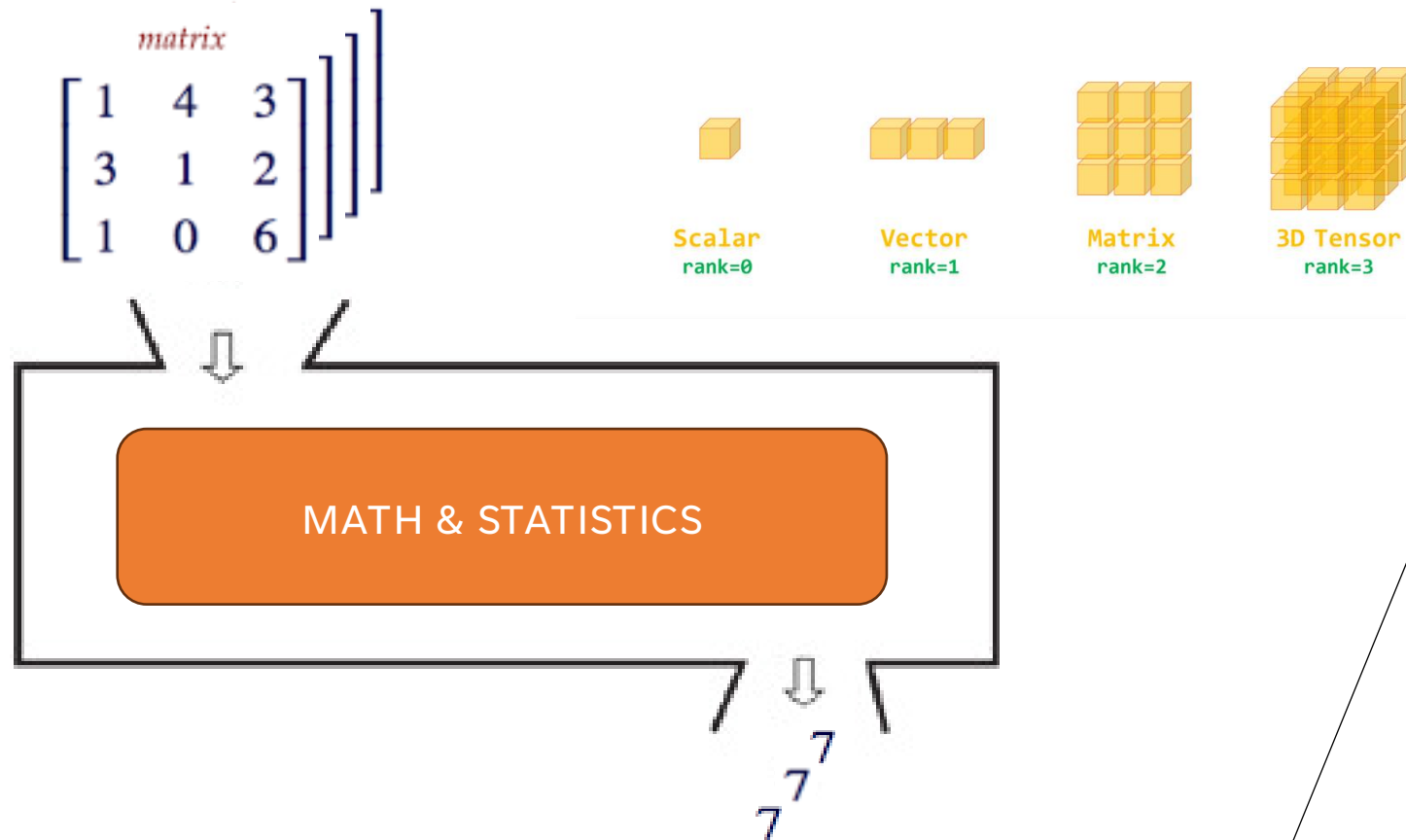
DEEP LEARNING

AI = Magic box = Function



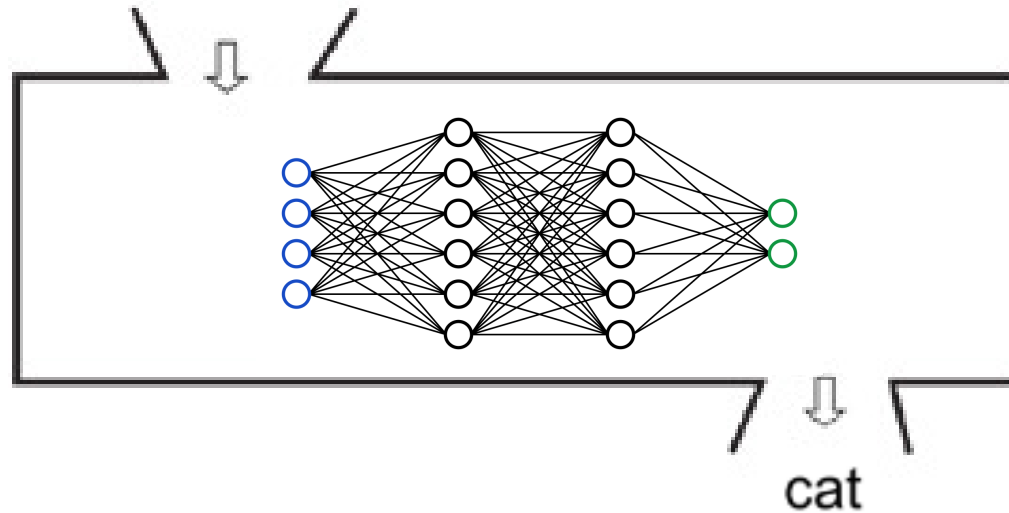
DEEP LEARNING

컴퓨터는 0과 1로 이루어져 있음 → input, output, 박스내부 모두 숫자로 구성

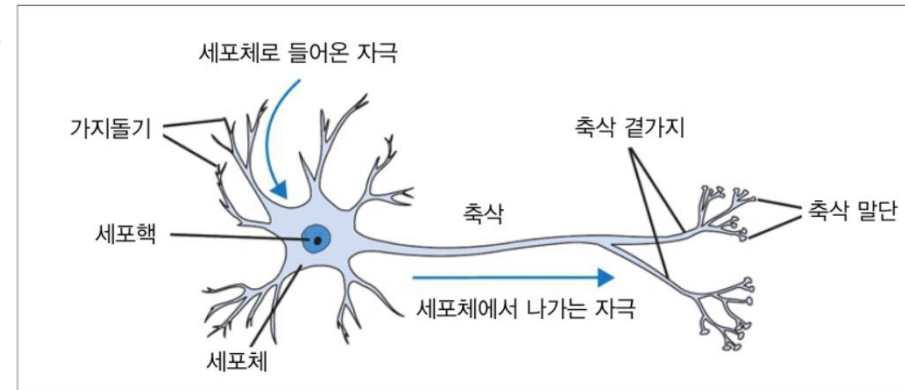


DEEP LEARNING

Box inside : Deep neural network

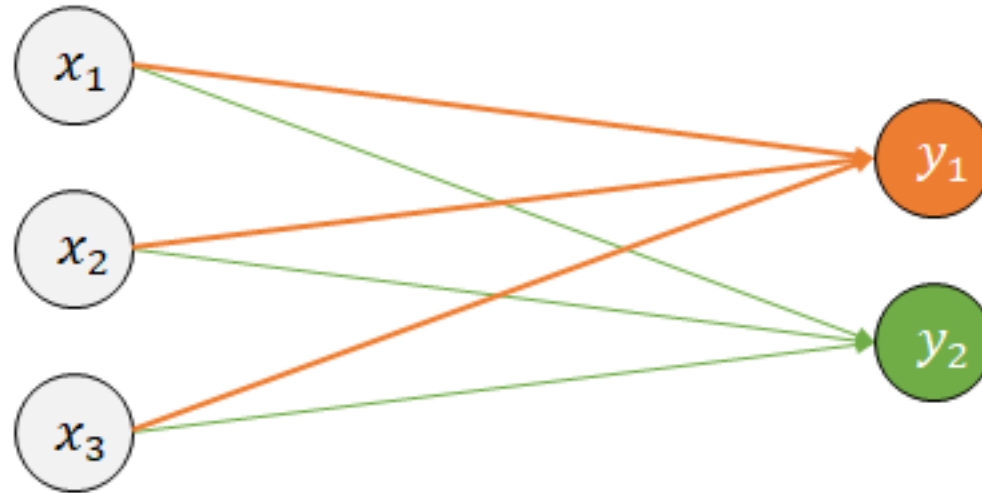


cf)



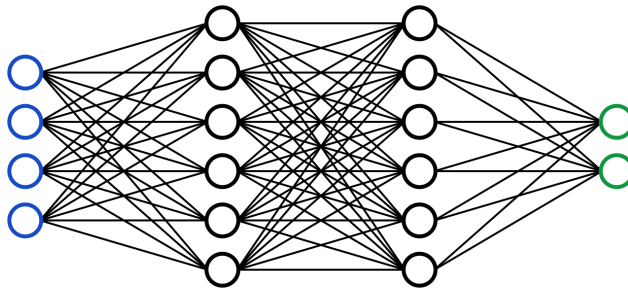
DEEP LEARNING

Neural network = Matrix multiplication



DEEP LEARNING

Neural network = Matrix multiplication + Non-linear transformation



Sigmoid	Tanh	RELU
$g(z) = \frac{1}{1 + e^{-z}}$	$g(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$	$g(z) = \max(0, z)$



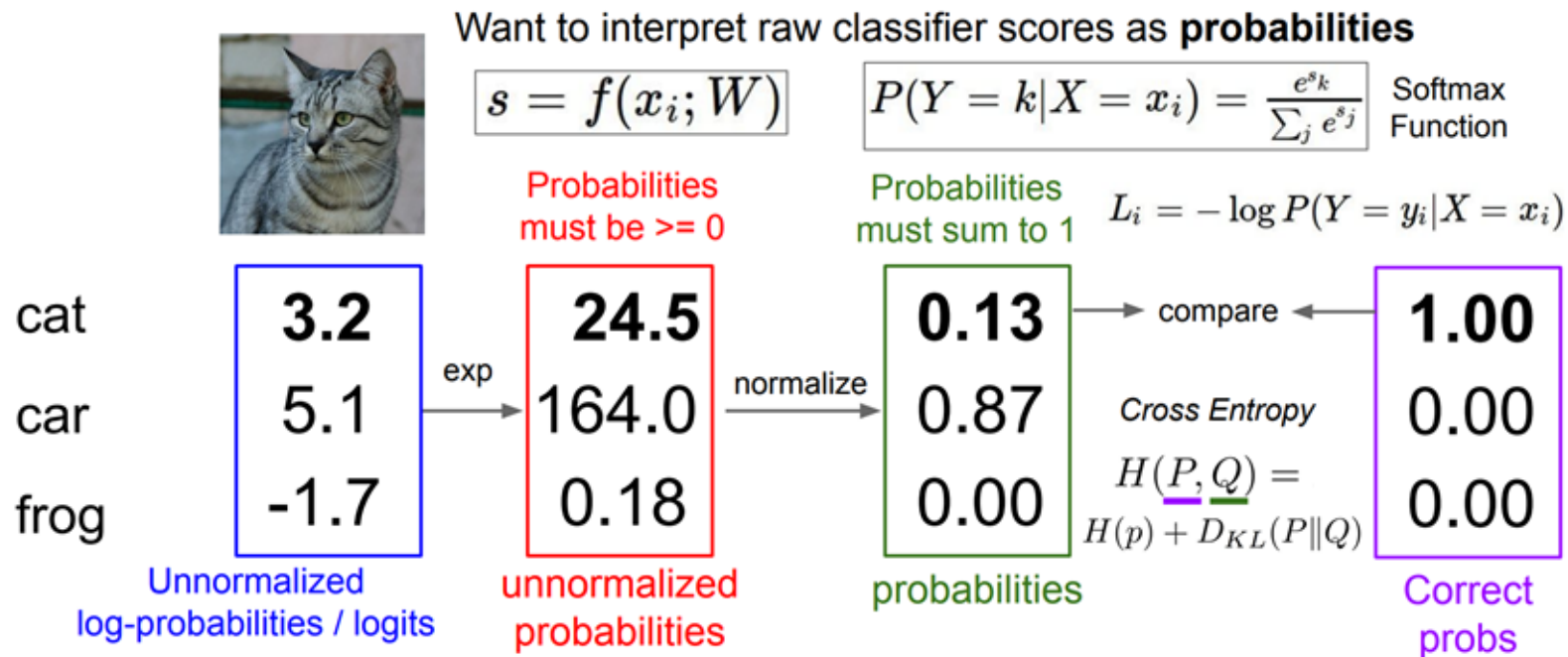
DEEP LEARNING

Neural network 내부의 가중치 w 들은 random 값으로 초기화 되어 있음

➔ 학습데이터를 통한 학습이 필요

1) Network의 예측값과 실제 정답 간의 차이(오차)를 정량화 (Error, Loss)

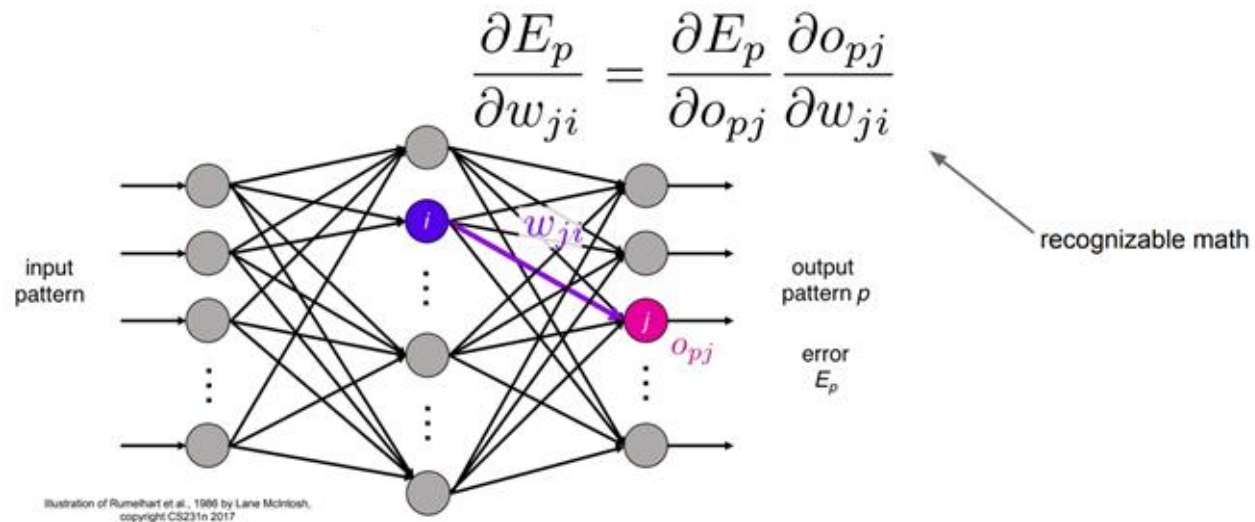
Softmax Classifier (Multinomial Logistic Regression)



DEEP LEARNING

2) Backpropagation, gradient descent 통한 가중치 w 들 update

Cf) gradient 는 PyTorch가 알아서 계산해줌



Q) $\frac{\partial E_p}{\partial w_{ji}} = 0.1$ 의 의미?

Cf) learning rate



DEEP LEARNING

3) GPU 안에 모든 학습 데이터를 다 넣는 것이 불가능

→ Mini-batch training 을 수행

- Optimizer – SGD or Adam
- Batch size
- Epoch



An abstract graphic design featuring two thin, dark gray lines that intersect on a light gray background. One line runs diagonally from the top-left towards the bottom-right, while the other runs from the top-right towards the bottom-left. The intersection point is located to the left of the text.

CONVOLUTIONAL NEURAL NETWORK

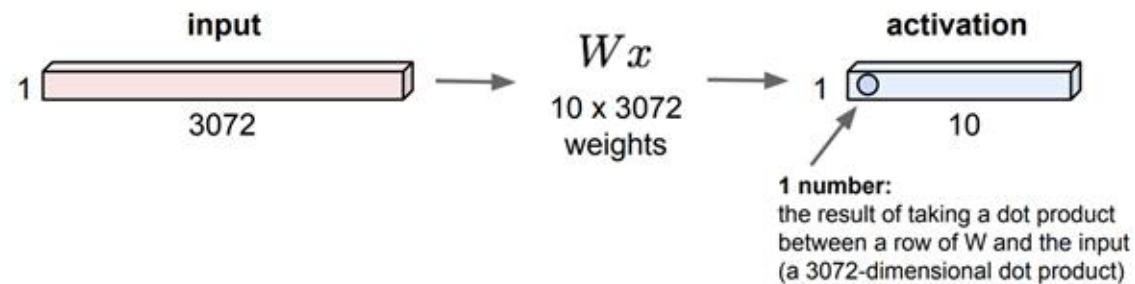
CONVOLUTIONAL NEURAL NETWORK

Multi-layer perceptron이 이미지 데이터를 처리할 때의 단점

- 1) 행렬 곱을 하기 위해 image tensor를 stretch 해야 함
➔ Spatial structure 소실됨

Fully Connected Layer

32x32x3 image -> stretch to 3072 x 1



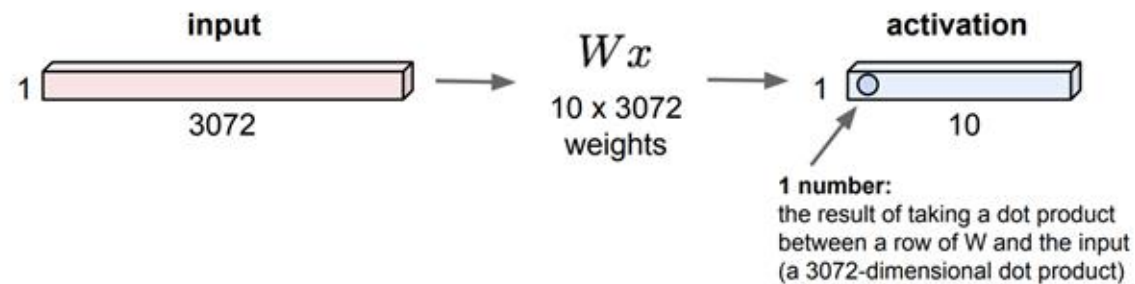
CONVOLUTIONAL NEURAL NETWORK

Multi-layer perceptron이 이미지 데이터를 처리할 때의 단점

- 2) 이미지의 크기가 커짐에 따라서
학습해야 하는 가중치 개수가 quadratic하게 증가

Fully Connected Layer

32x32x3 image -> stretch to 3072 x 1

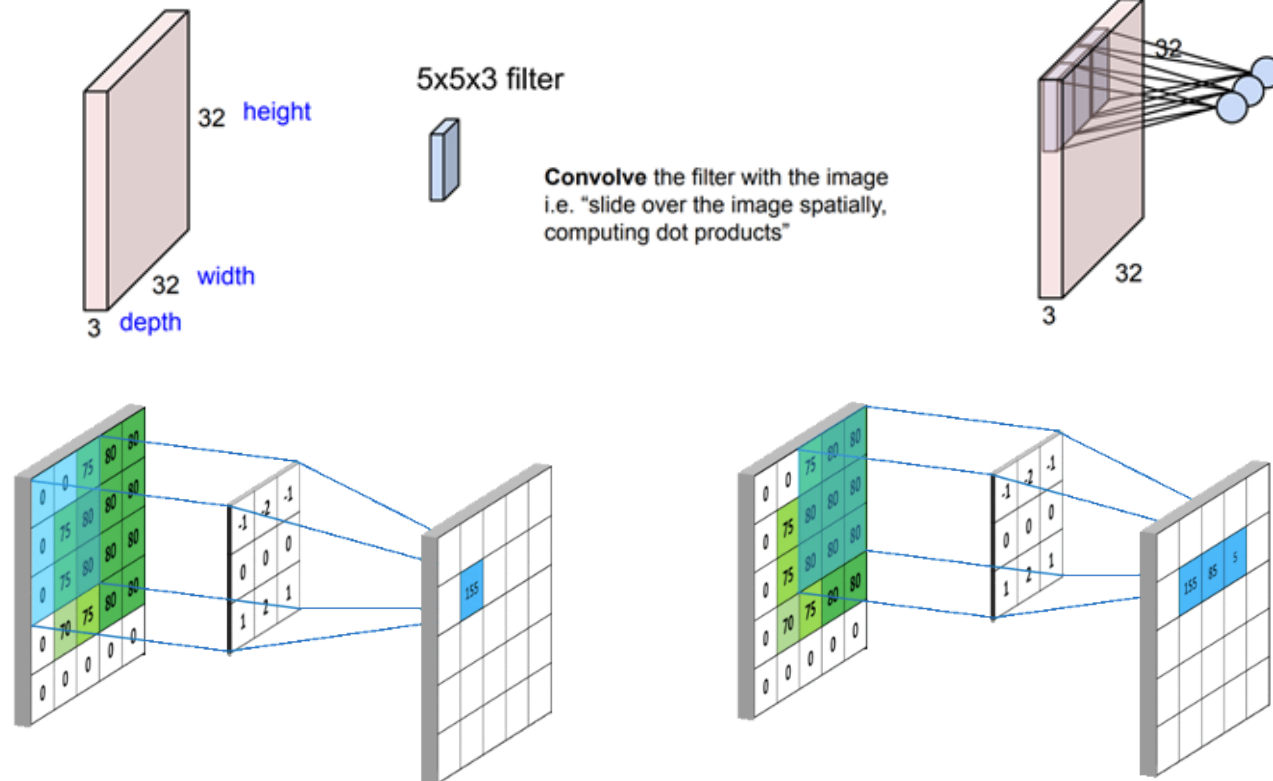


CONVOLUTIONAL NEURAL NETWORK

Convolution layer의 등장

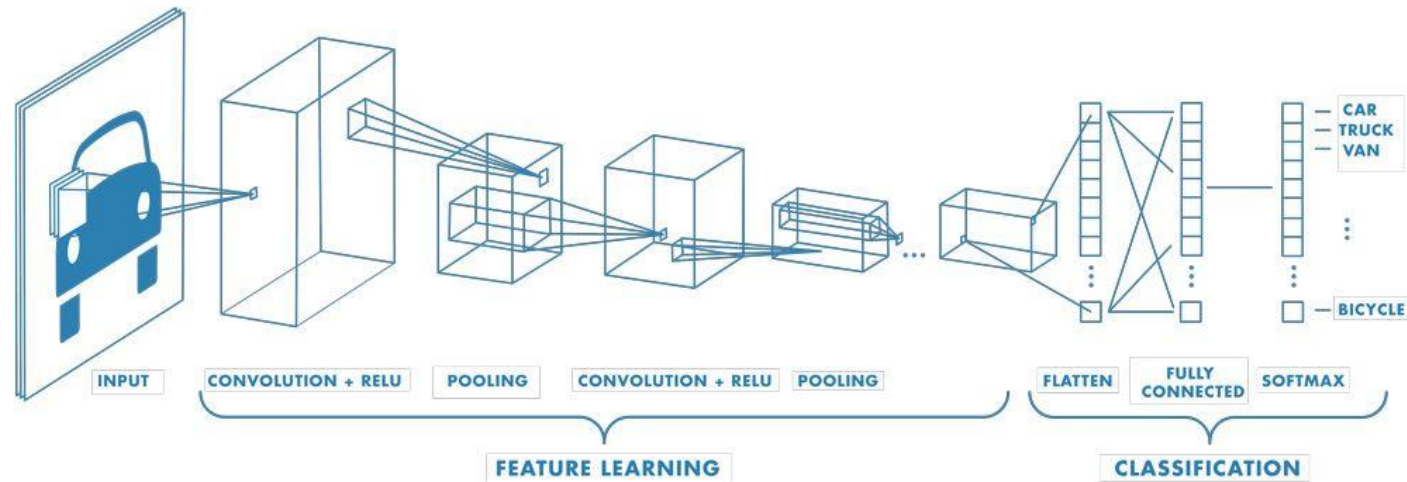
Convolution Layer

32x32x3 image -> preserve spatial structure



CONVOLUTIONAL NEURAL NETWORK

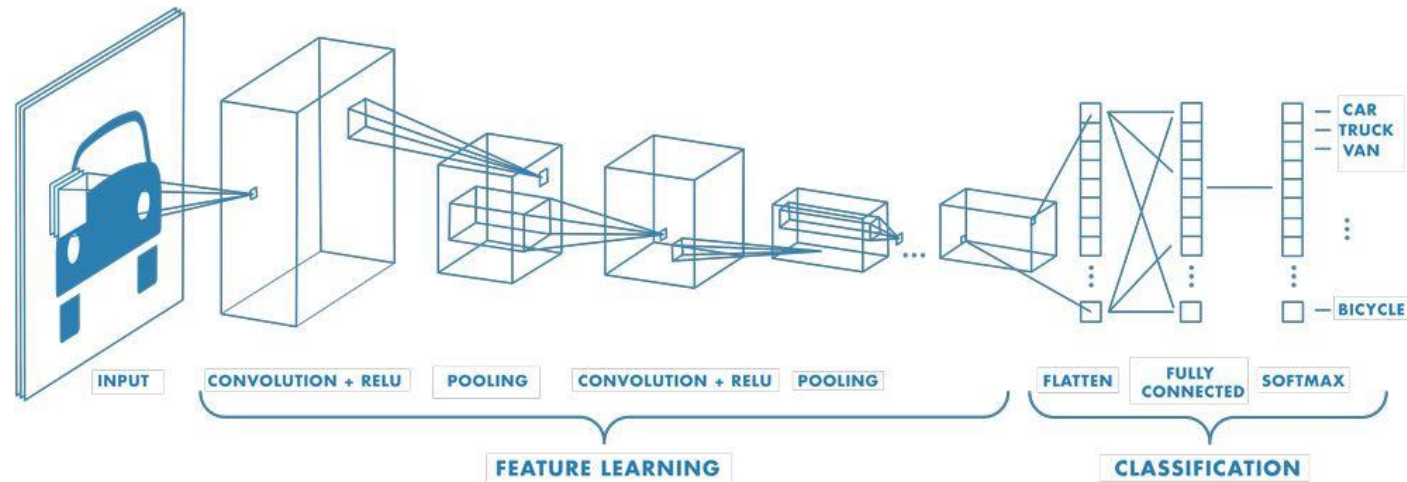
CNN의 전체 구조 : [CONV & RELU & POOL] x N → Flatten → FC → Softmax



CONVOLUTIONAL NEURAL NETWORK

CNN의 이점 :

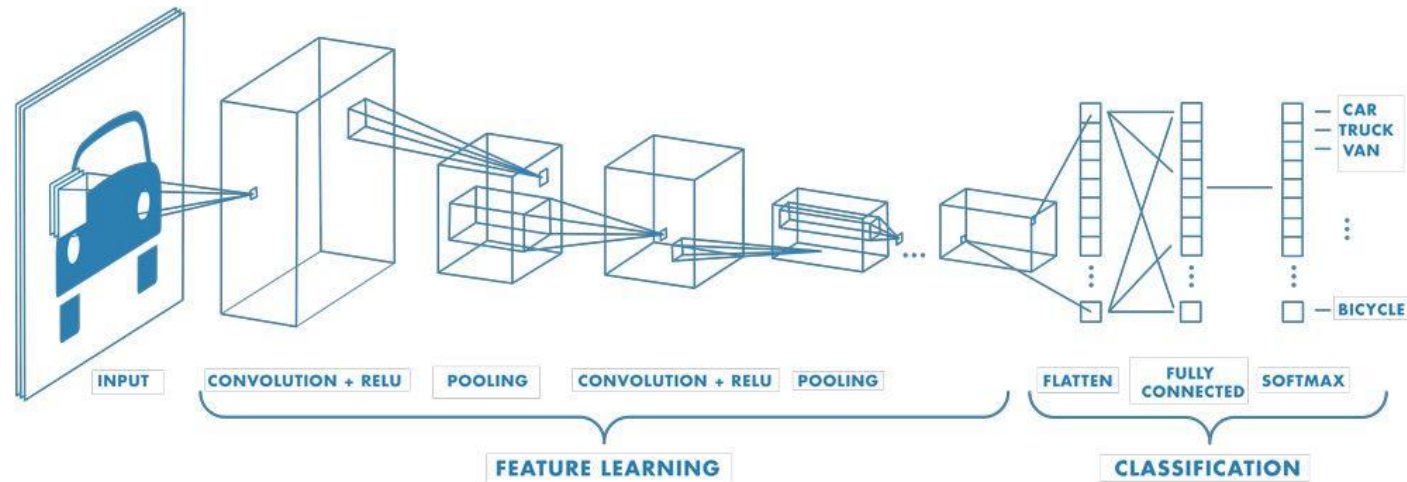
1) 공간적 구조(Spatial structure)를 preserve하면서 feature learning이 가능



CONVOLUTIONAL NEURAL NETWORK

CNN의 이점 :

2) 가중치 공유를 통한 학습해야 할 파라미터 개수 감소

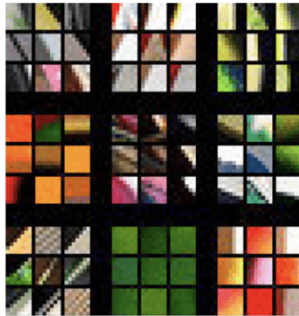


CONVOLUTIONAL NEURAL NETWORK

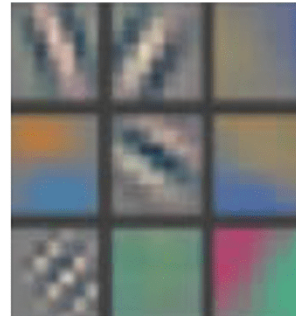
CNN의 이점 :

3) 계층적 특징 학습이 가능

Patches from Input Image



Layer 1 Structure



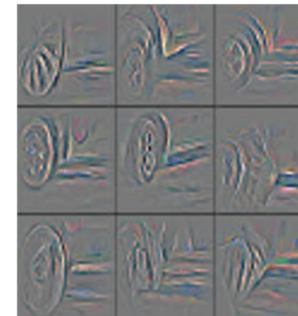
The first hidden layer learns to identify basic structural elements such as edges and color blobs

Patches from Input Image



CNNs learn hierarchical structure after several layers

Layer 5 Structure



CONVOLUTIONAL NEURAL NETWORK

CNN의 이점 :

4) 위치 변화에도 일관된 특징 학습/추론이 가능



CNN EXAMPLE – 1) VGGNET

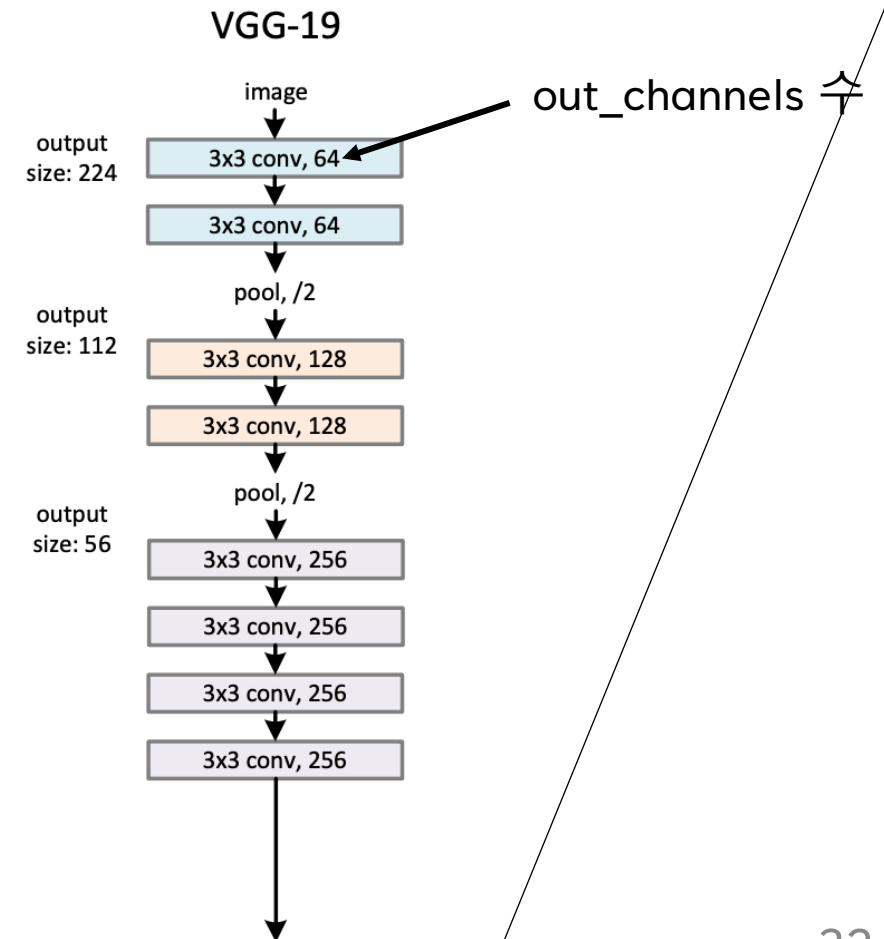
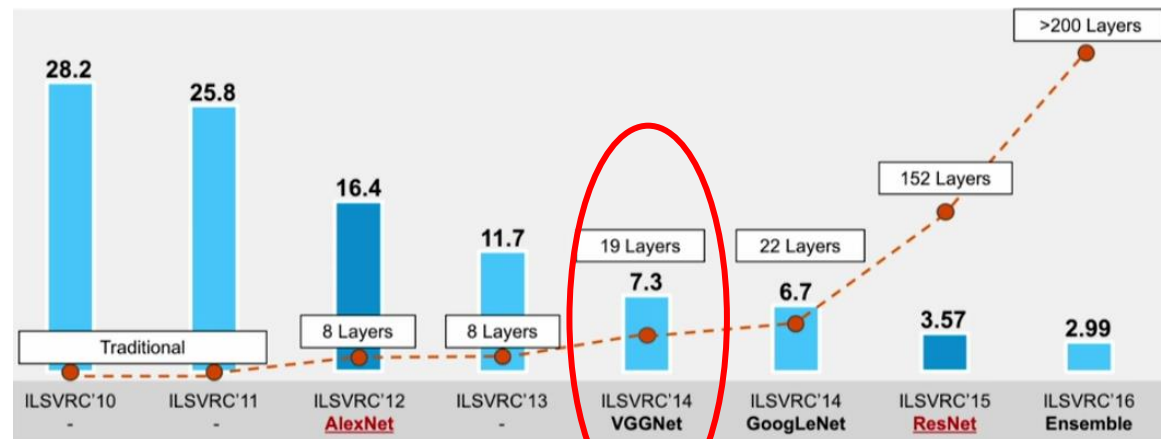
2014년 ImageNet 대회 2등 수상 – 옥스포드대 VGG 연구팀

“Smaller filters, Deeper networks”

3x3 kernel (& padding 1) convolution layer만 사용

네트워크 깊이 : 8 layers → 19 layers

오차율 : 11.7 % → 7.3 %



CNN EXAMPLE – 2) RESNET

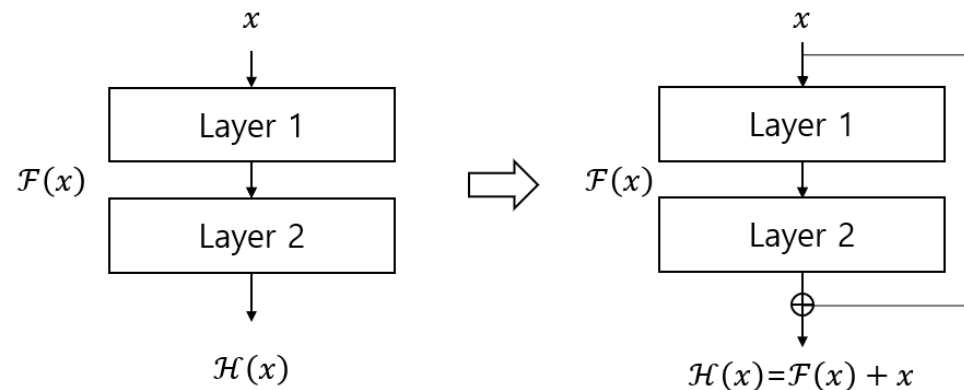
2015년 ImageNet 대회 1등 수상 - Microsoft
RESidual NETwork의 약자

네트워크 layer를 깊게 쌓을 수록 optimize가 어려워짐

→ Short connection (identity mapping)을 추가

→ 각 block에서는 잔차 ($H(x) - x = F(x)$)만 학습하면 됨

→ 네트워크 Optimize가 용이해짐

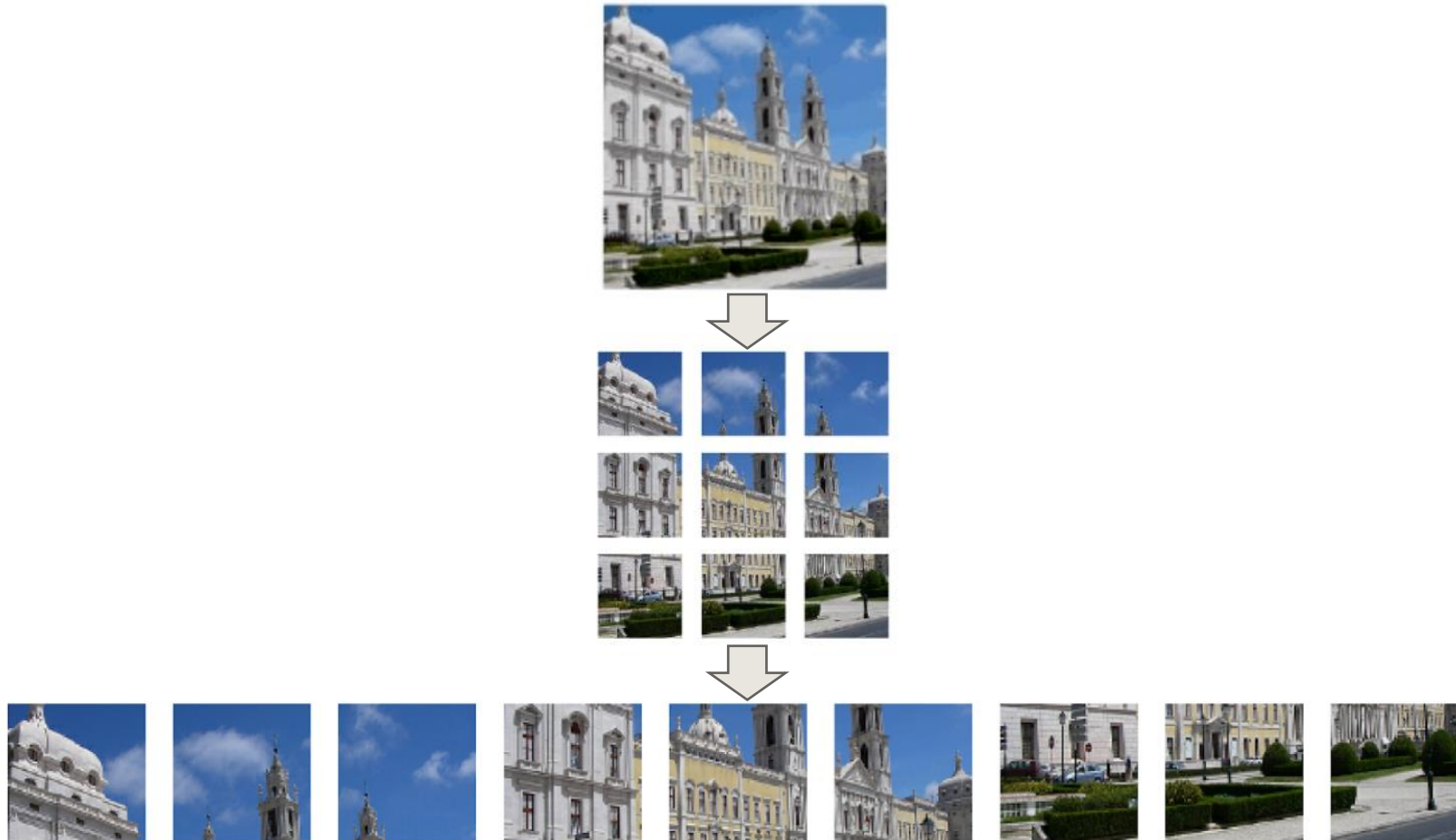




VISION TRANSFORMER

VISION TRANSFORMER

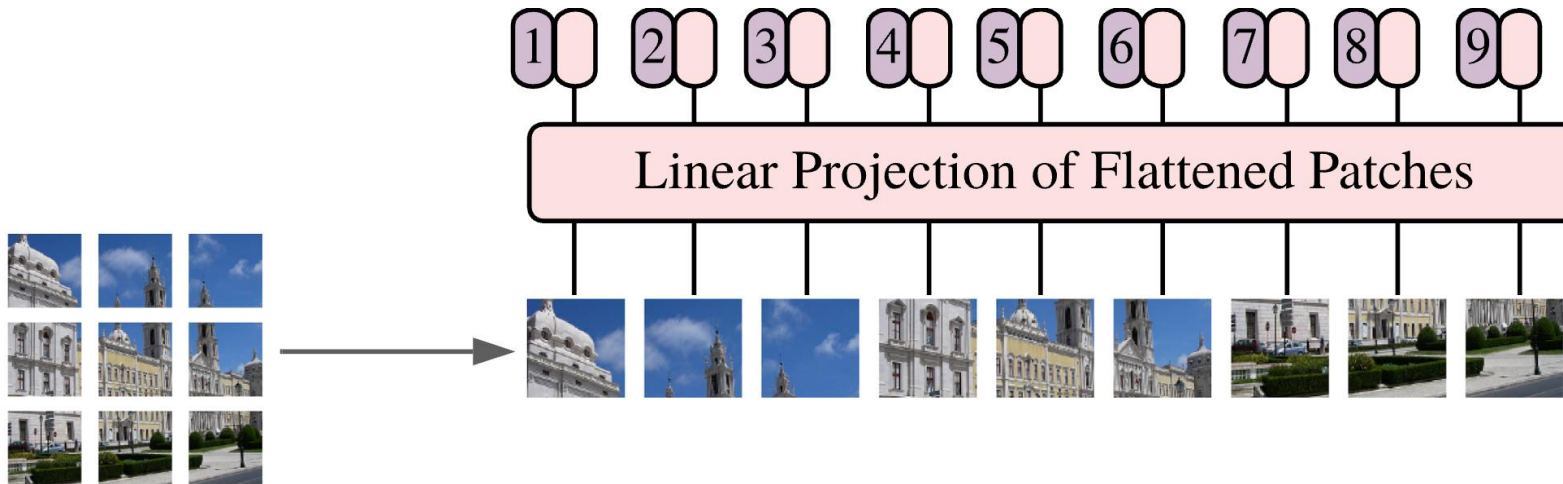
이미지를 패치/타일 단위로 쪼개면 Sequence로 생각 가능



VISION TRANSFORMER

각 패치를 Flatten한 후 Linear projection하여 Embedding으로 표현

+ Positional encoding

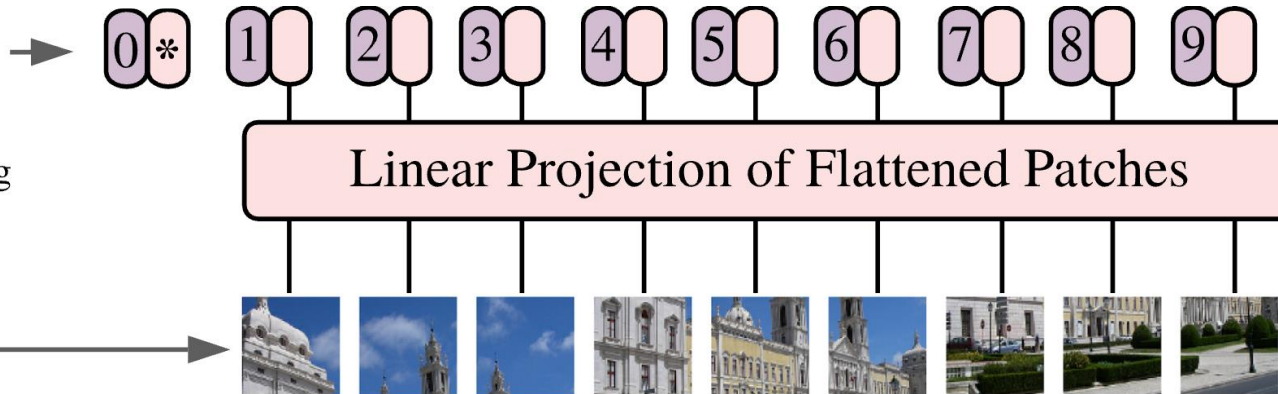


VISION TRANSFORMER

Extra learnable [class] embedding을 sequence 맨 앞에 Concatenate
(이미지 전체에 대한 context를 담기 위해)

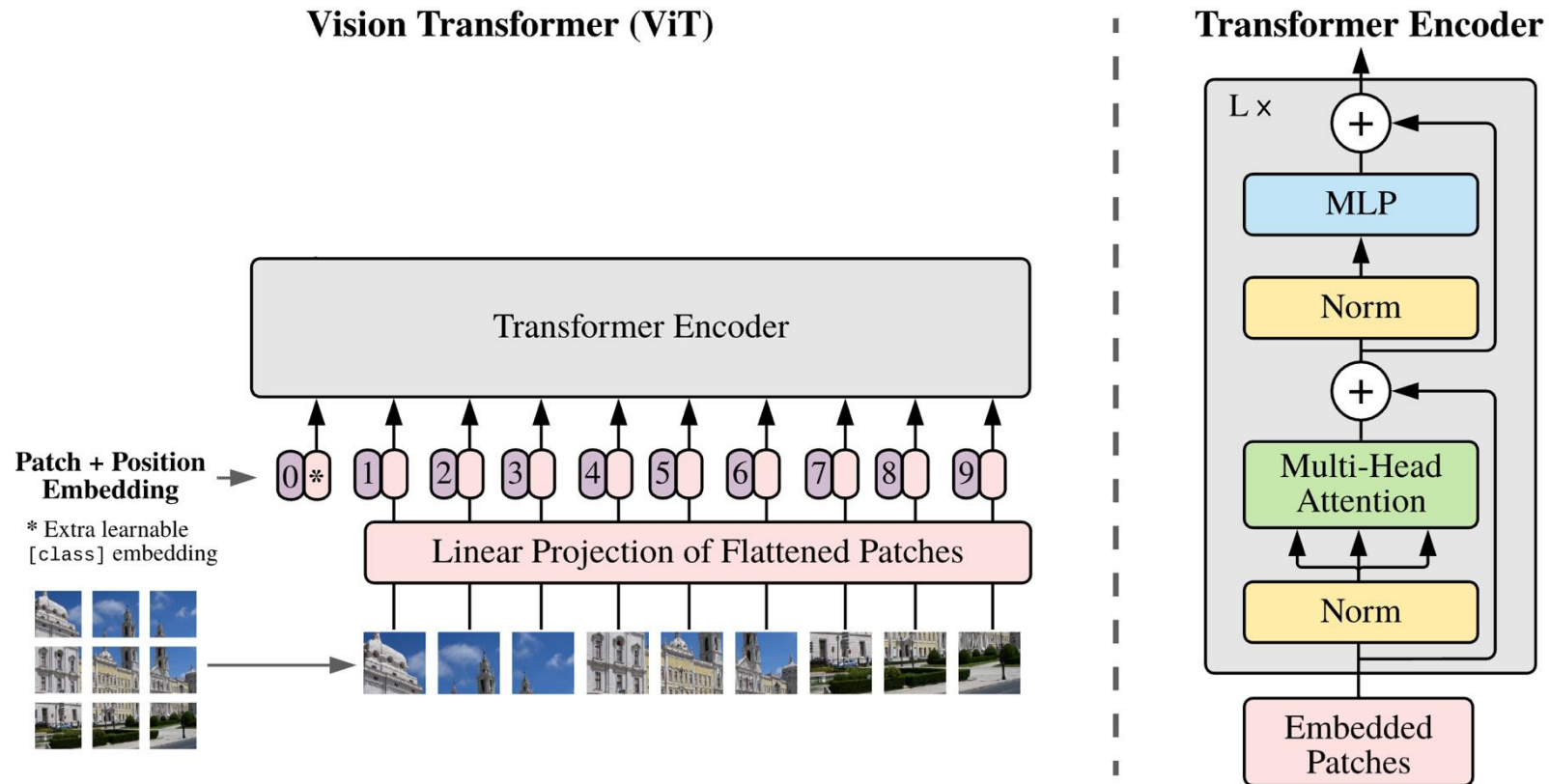
Patch + Position
Embedding

* Extra learnable
[class] embedding



VISION TRANSFORMER

Transformer encoder (self-attention blocks)를 통해 패치간 관계성 모델링



MECHANISM OF SELF-ATTENTION

Slide credit :

https://www.youtube.com/watch?v=4naXLhVfeho&ab_channel=NeuralBreakdownwithAVB

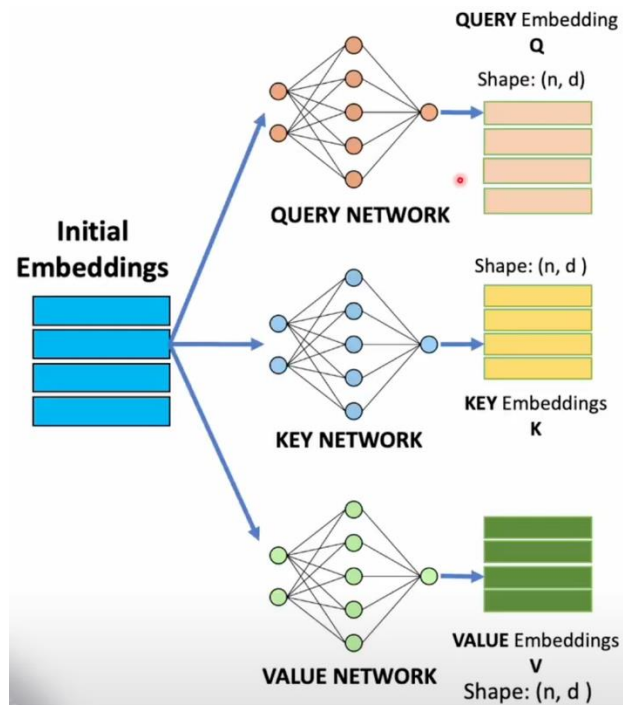
Initial
Embeddings



MECHANISM OF SELF-ATTENTION

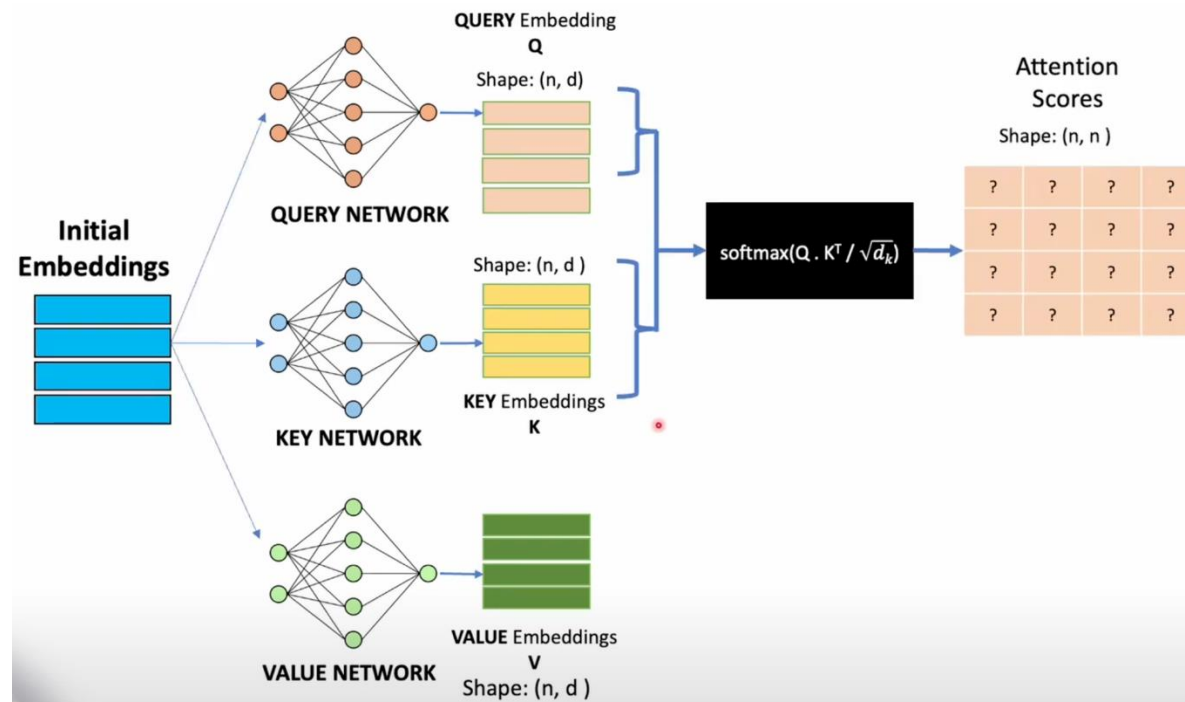
(1) Projection을 통해 Q/K/V 역할 분리

Query/Key/Value network : 일반적으로 single linear layer



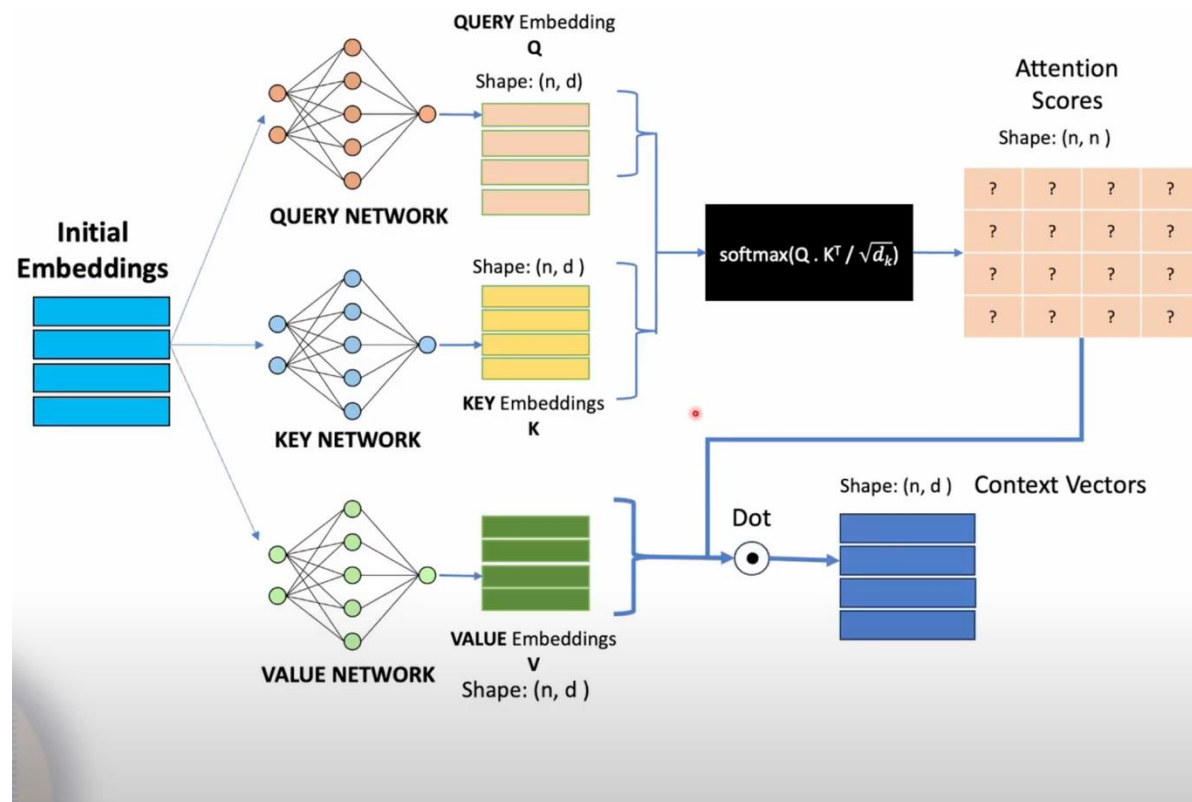
MECHANISM OF SELF-ATTENTION

(2) Attention score (alignment score) 계산



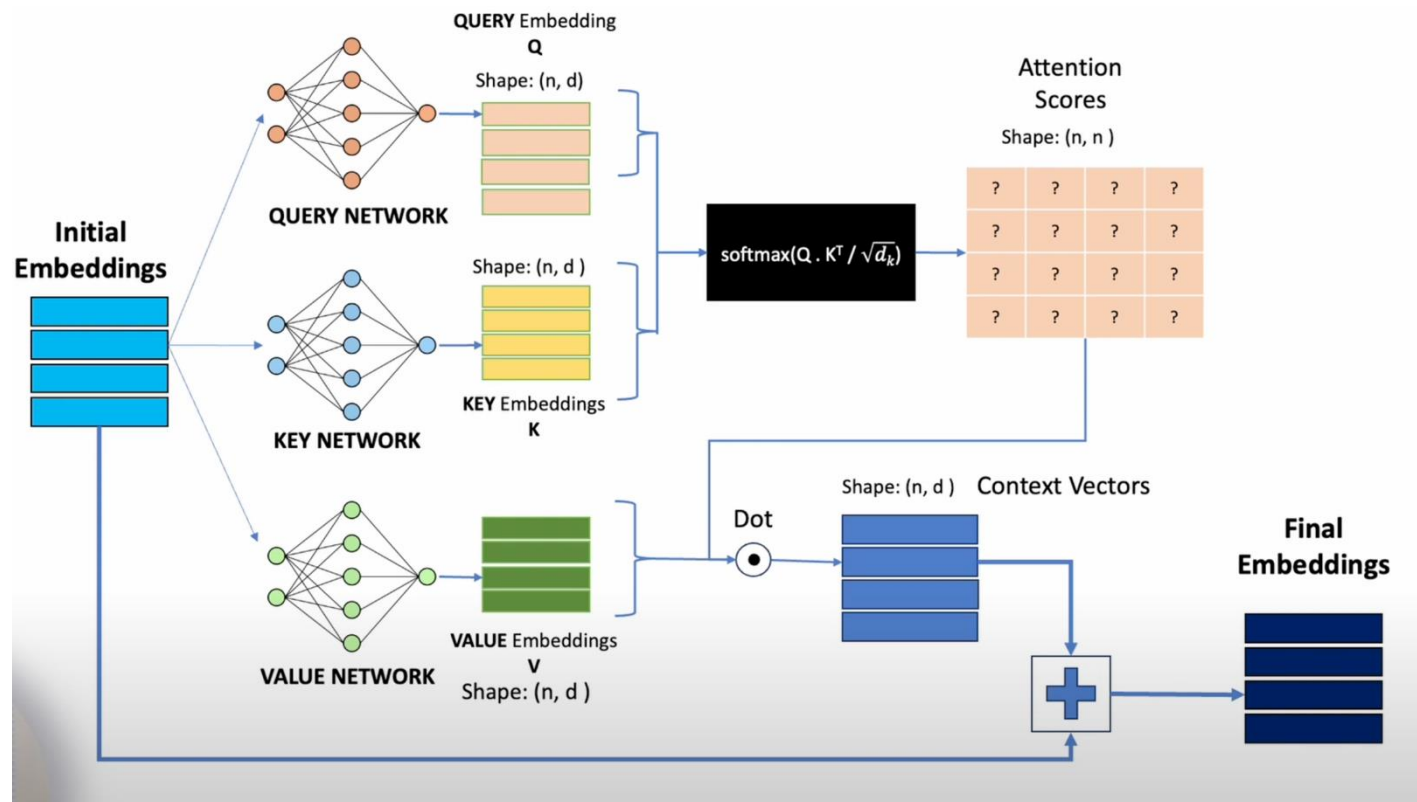
MECHANISM OF SELF-ATTENTION

(3) Context vector 계산



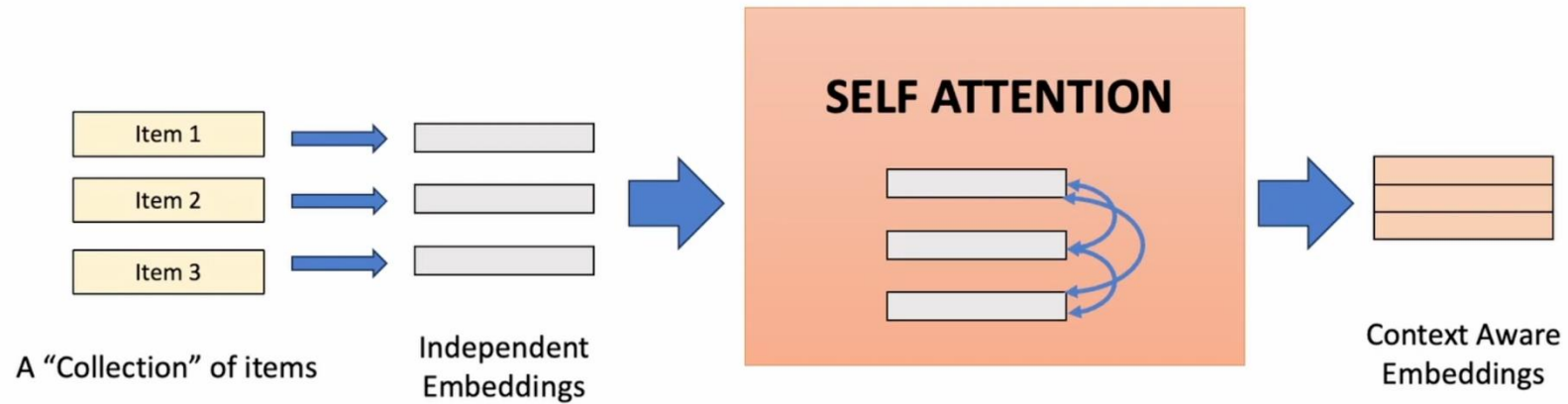
MECHANISM OF SELF-ATTENTION

(4) 최종 Embedding 계산



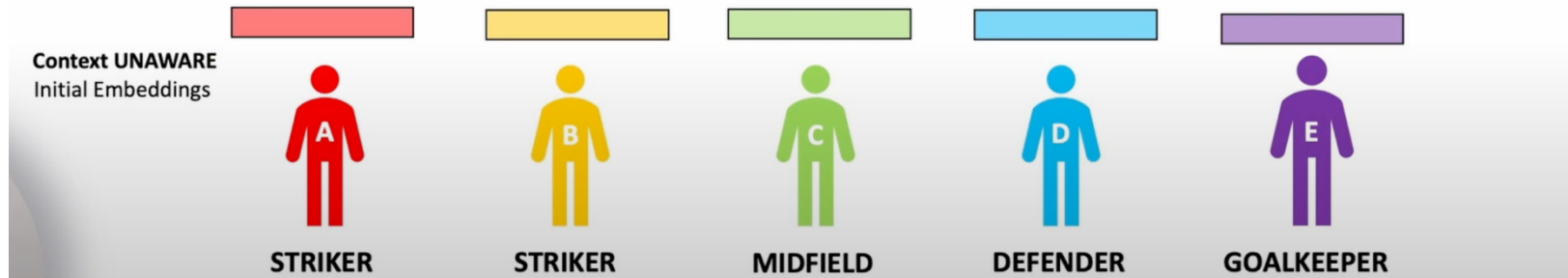
ROLE OF SELF-ATTENTION

“Independent” embeddings → “Context-aware” embeddings



ROLE OF SELF-ATTENTION

Ex) 축구



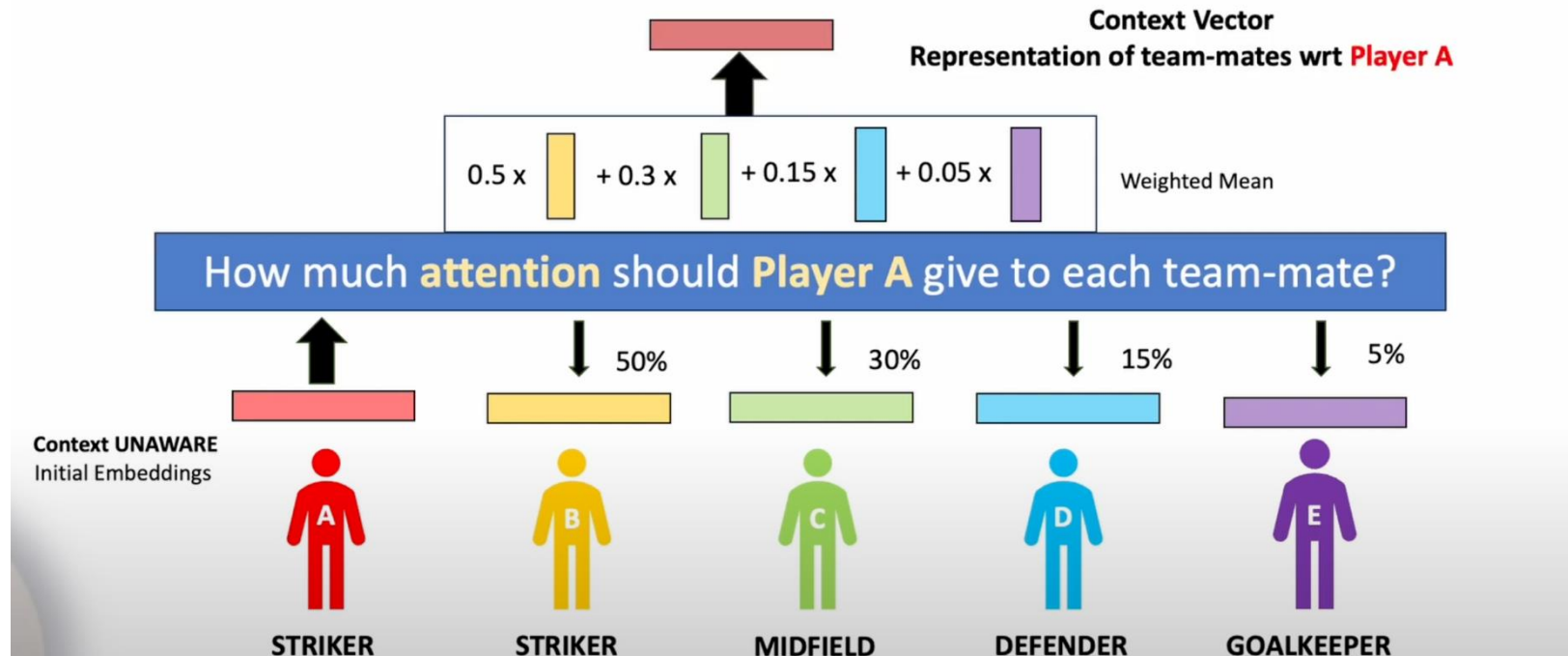
ROLE OF SELF-ATTENTION

Ex) 축구



ROLE OF SELF-ATTENTION

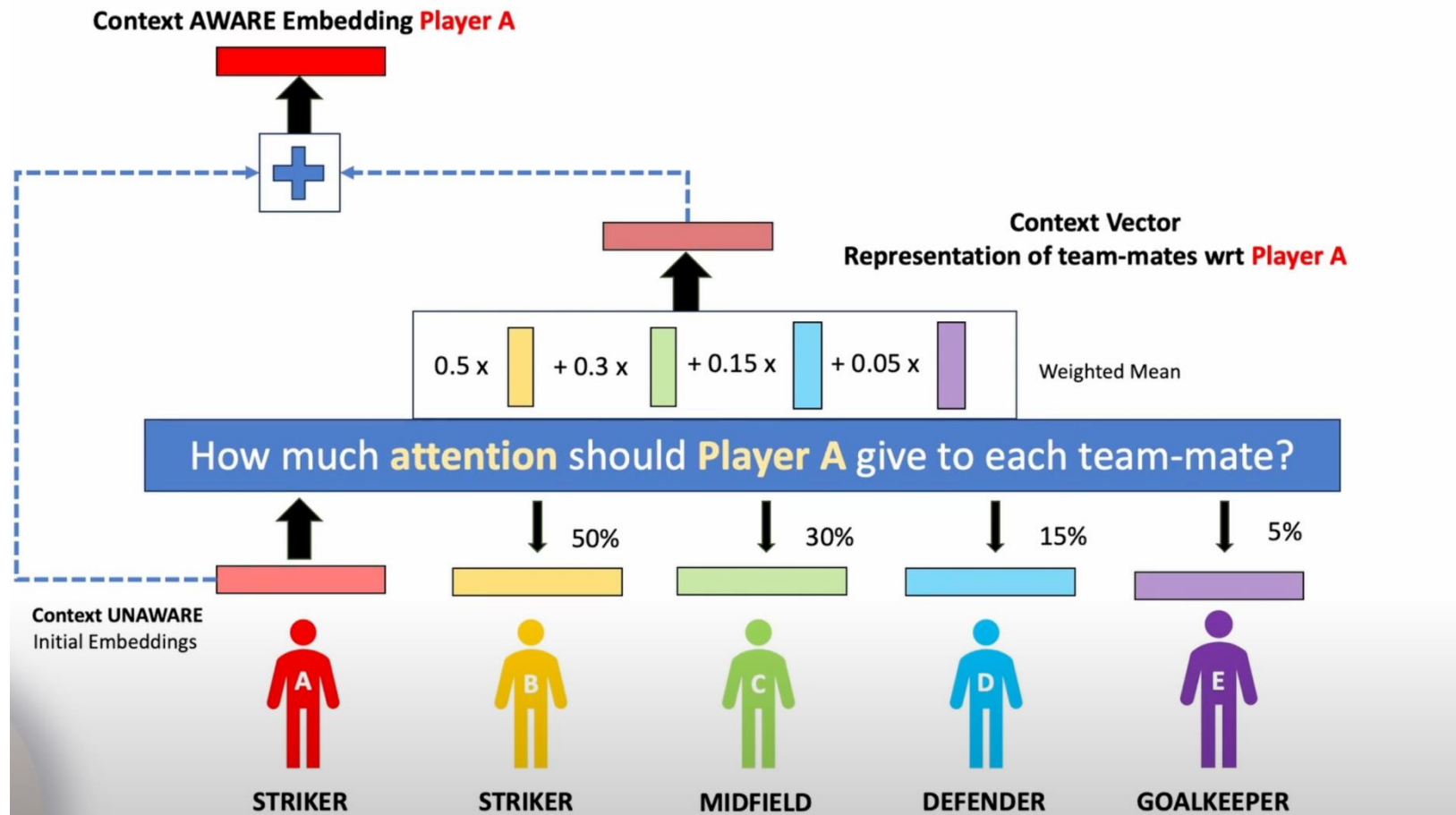
Ex) 축구



ROLE OF SELF-ATTENTION

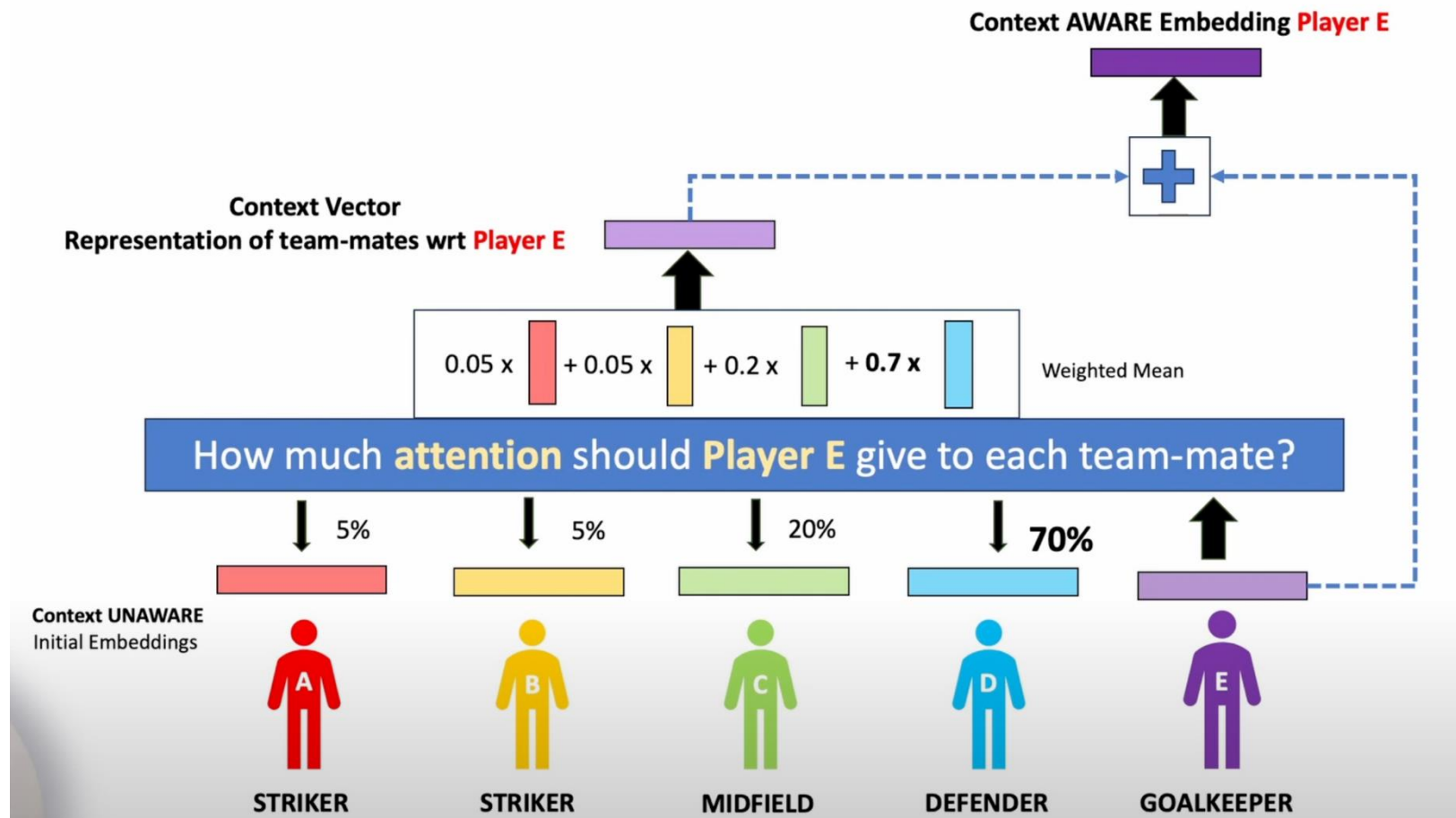
Ex) 축구

Context Unaware + Team Context = Context Aware



ROLE OF SELF-ATTENTION

Ex) 축구



ROLE OF SELF-ATTENTION

Ex) 축구

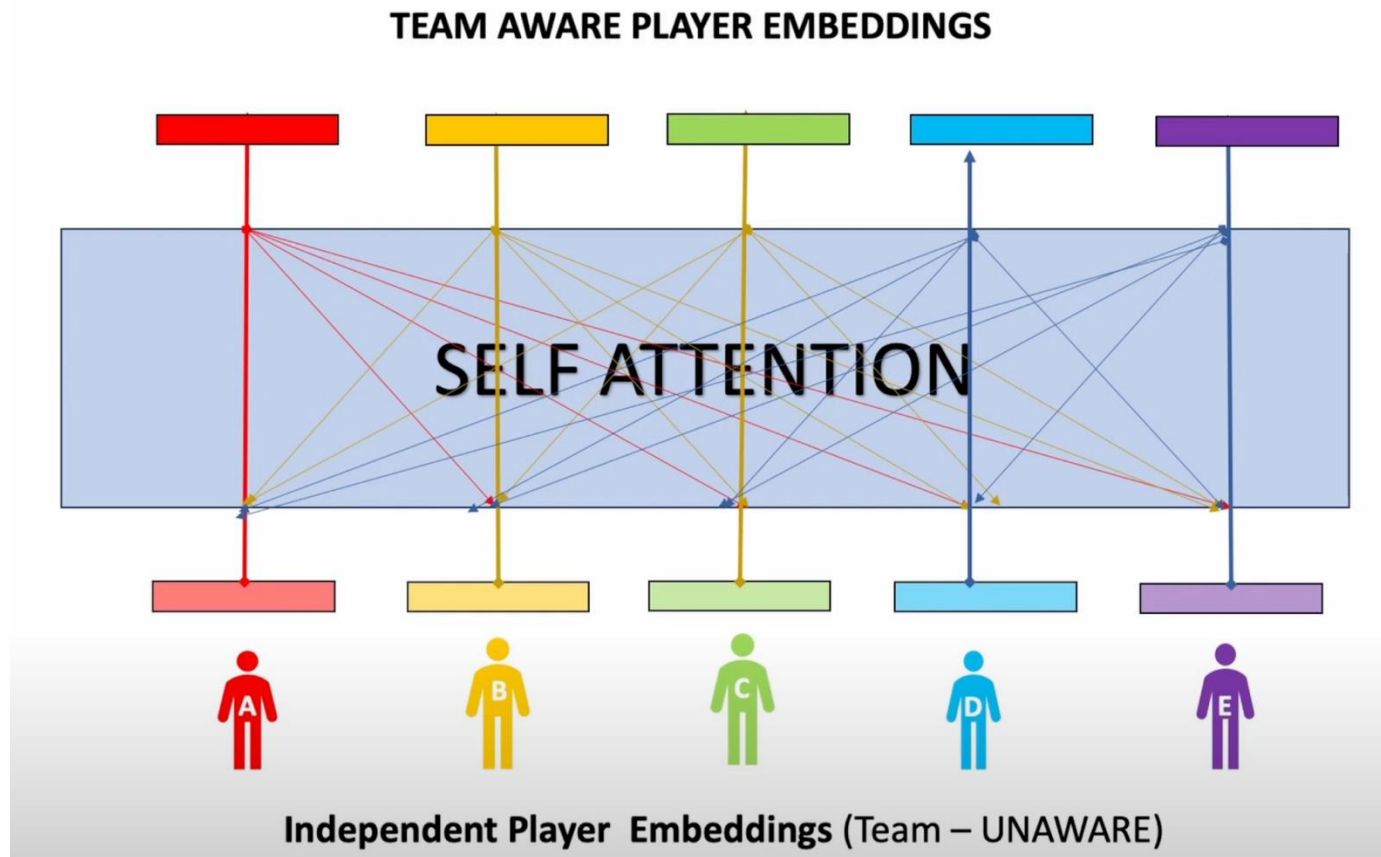


The Attention Matrix tells us how much attention each player gives to its team mates.



ROLE OF SELF-ATTENTION

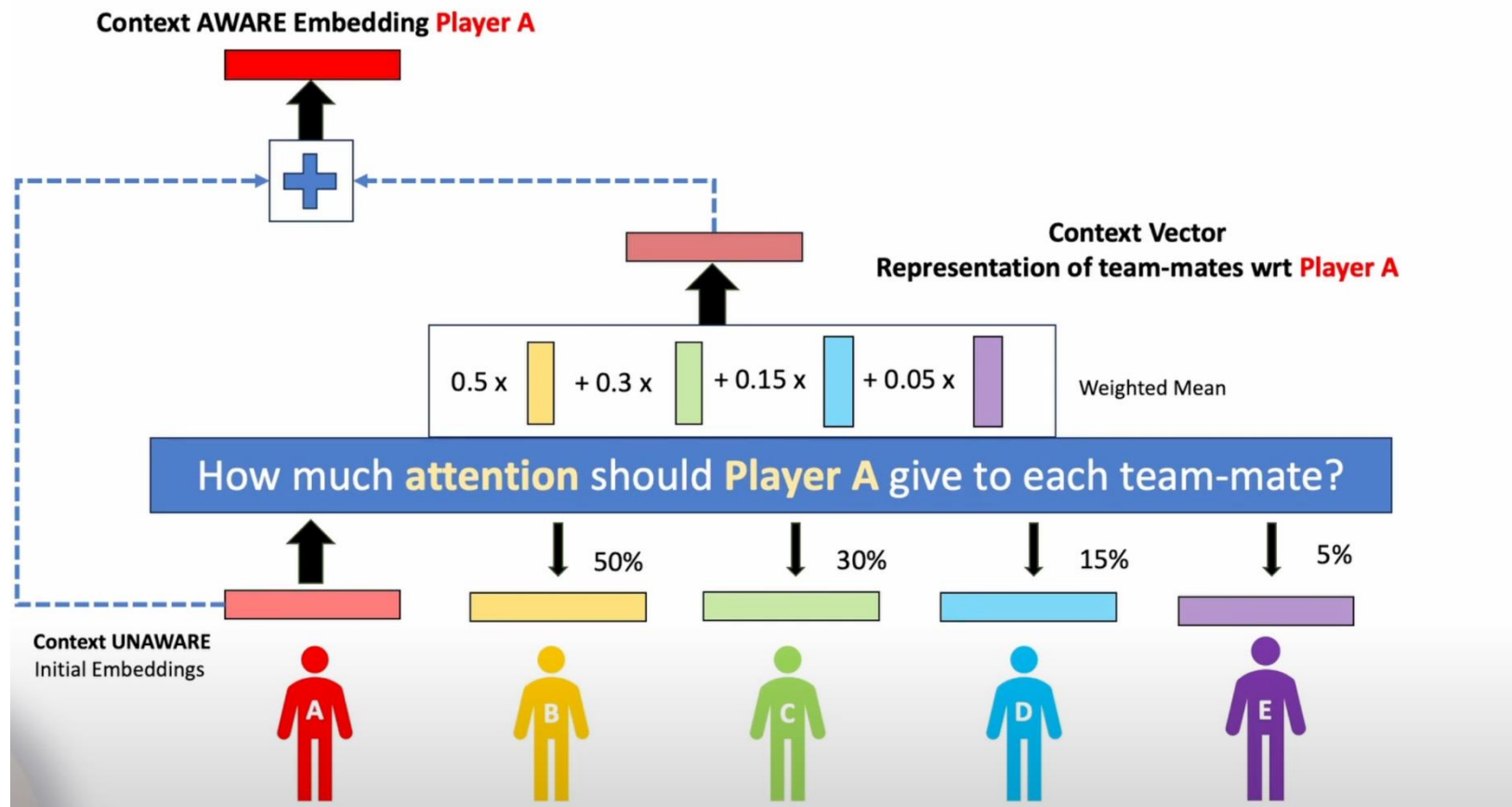
Ex) 축구



ROLE OF SELF-ATTENTION

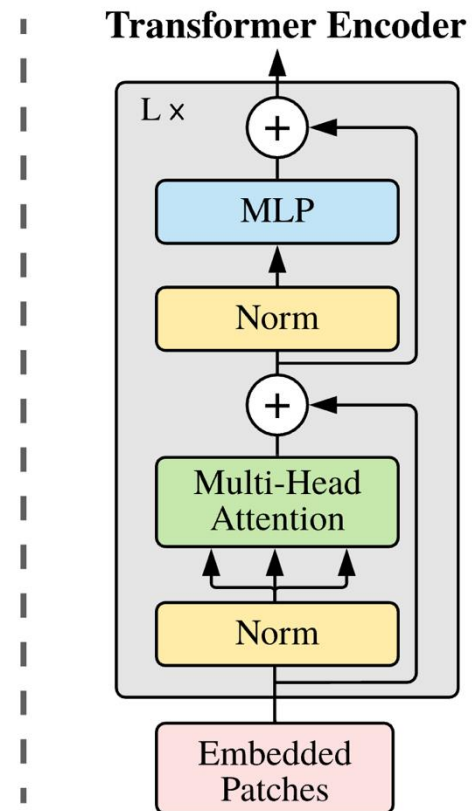
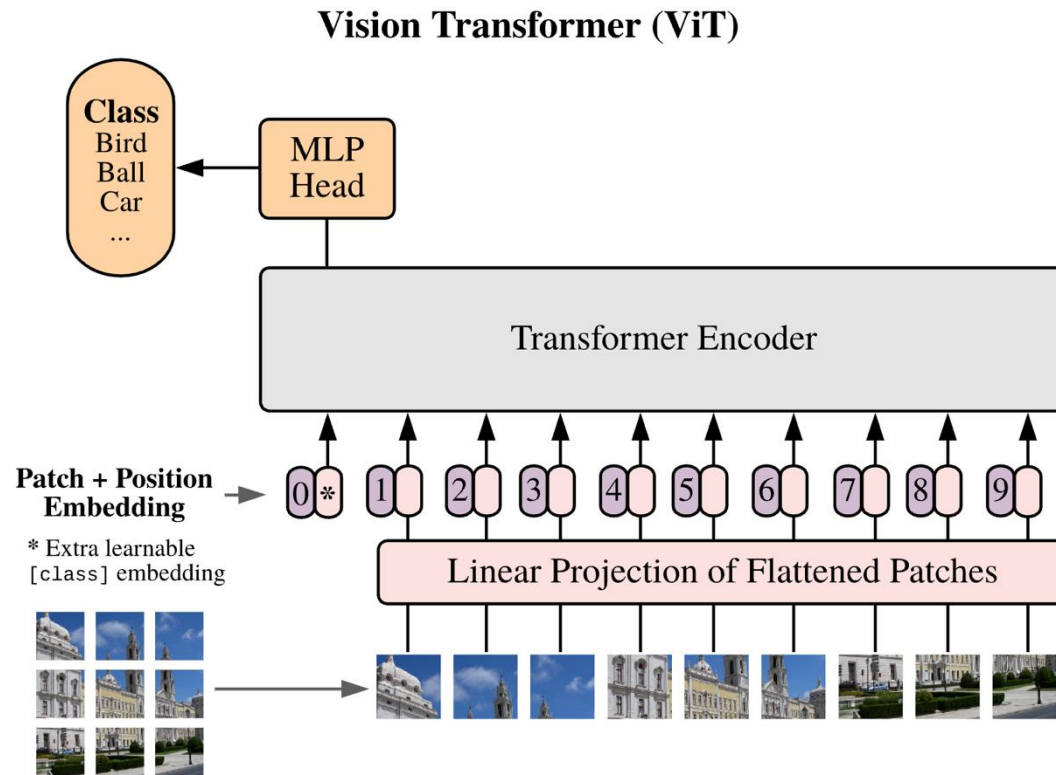
Ex2) LOL

Context Unaware + Team Context = Context Aware



VISION TRANSFORMER

0번째 position에 대한 최종 embedding에 대해 classification 수행



An abstract graphic on a light gray background. Two thin, dark gray lines intersect. One line is nearly vertical, starting from the top center and extending towards the bottom right. The other line is nearly horizontal, starting from the top left and extending towards the middle right. The intersection point is located in the upper-left quadrant of the image. To the right of the intersection, the word "SUMMARY" is written in a bold, black, sans-serif font.

SUMMARY

SUMMARY

1. Perception – overview
2. Image classification
3. Convolutional neural network
4. Vision transformer



NEXT LECTURE

Perception (2) – 2D object detection



Two thin, dark gray lines intersect diagonally on a light gray background. One line slopes downwards from left to right, and the other slopes upwards from left to right.

QUESTIONS?